SMART INDIA HACKATHON

WORKSHOP • PROPOSED SOLUTION

FarmLink Direct

An AI-Assisted Digital Marketplace for Fairer Farm Prices and Affordable Produce

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Date	23/10/2026
Problem ID	SIH 26033

Problem Title
Multiple intermediaries reduce farmers' earnings and increase consumer prices.

Problem Creator's Organization
Ministry of Consumer Affairs, Food & Public Distribution

Theme
Agriculture, FoodTech & Rural Development

Concept summary: FarmLink Direct is a proposed online marketplace that helps farmers and Farmer Producer Organisations (FPOs) list produce, compare buyer offers, arrange shared transport, and use demand and route predictions to reduce avoidable costs. The concept is designed to complement existing agricultural market systems rather than replace necessary quality checks or regulations.

1. Proposed Solution

FarmLink Direct is a mobile-friendly web platform connecting farmers, FPOs, households, retailers, hotels, restaurants, and bulk buyers through one transparent marketplace. Farmers can publish available produce, quantity, harvest date, quality grade, expected price, and pickup location. Buyers can compare listings and place orders directly.

Key features

- **Direct digital listings:** Farmers and FPOs can upload produce details, photographs, quantity, and a preferred price.
- **Buyer matching:** Consumers and bulk buyers can search by crop, location, quantity, quality, and delivery date.
- **Transparent price comparison:** The platform displays seller prices, estimated delivery charges, quality details, and order totals before checkout.
- **Logistics coordination:** Nearby orders can be grouped and matched with available local transport providers to reduce empty trips and per-order transport costs.
- **AI demand forecasting:** Historical orders, seasonal patterns, local events, and available market information can estimate likely demand for selected crops.
- **Route optimization:** A routing module can suggest efficient delivery sequences using distance, vehicle capacity, delivery windows, and road conditions where available.
- **Trust and support:** Order tracking, digital receipts, ratings, complaint handling, and optional assisted registration through FPOs or local help centres.

How it addresses the problem

By allowing buyers to discover produce from farmers and FPOs directly, the platform can reduce reliance on unnecessary layers of resale. Price and quality information helps both sides make informed decisions. Grouped deliveries and route planning can reduce logistics waste. Demand forecasts may help farmers and FPOs plan harvest collection and marketing more effectively. The platform cannot eliminate every intermediary: transporters, packers, quality assessors, and regulated market services may still be useful. Its goal is to make each service visible, fairly priced, and accountable.

Innovation and uniqueness

- A combined marketplace, shared-logistics planner, and demand forecasting dashboard in one workflow.
- A low-bandwidth, mobile-first design with local-language labels and an assisted mode for users who are less comfortable with digital tools.
- Bulk-order aggregation so small farmers can combine compatible quantities through an FPO without losing individual lot-level records.
- A transparent cost breakdown showing produce price, platform fee (if any), packing, transport, and final buyer price.

2. Technical Approach

Suggested technology stack

Layer	Proposed technology	Purpose
Frontend	HTML, CSS, JavaScript; responsive UI	Farmer, buyer, and logistics screens
Backend	Python with Django or Django REST Framework	Accounts, listings, orders, and APIs
Database	PostgreSQL	Users, inventory, orders, and price history
AI/analytics	Python, pandas, scikit-learn	Baseline demand prediction and evaluation
Maps/routing	OpenStreetMap-based mapping and routing service	Distance estimation and delivery sequencing
Notifications	Email/SMS or in-app alerts, subject to availability	Order and delivery updates

These are proposed technologies for a prototype. Final selection will depend on team skills, cost, connectivity, data availability, and integration requirements.

Implementation Methodology

Prototype workflow

1. Farmer/FPO registers	2. Lists produce & quantity	3. Buyer searches & places order	4. Payment/order confirmation	5. Logistics grouping & route	6. Delivery and feedback

AI workflow: collect permitted historical order and seasonal data → clean and aggregate the data → create a simple baseline forecast → compare predicted demand with actual demand → improve the model after sufficient data is collected. For routing, gather delivery stops and vehicle constraints → calculate feasible routes → compare distance/time estimates → assign grouped orders → update routes when an order or road condition changes.

Phased implementation

- **Phase 1 — Discovery:** Interview a small group of farmers, FPO representatives, retailers, consumers, and transport providers. Identify the first crops and service area.
- **Phase 2 — MVP:** Build registration, produce listings, search/filter, order placement, order status, and basic admin tools.
- **Phase 3 — Logistics pilot:** Add transport-provider records, delivery grouping, estimated delivery costs, and route suggestions.
- **Phase 4 — AI pilot:** Use sample or consented historical data to test a simple demand forecast; show confidence/limitations and allow human review.
- **Phase 5 — Evaluation:** Pilot with a limited number of users, measure results, fix issues, and expand only after feedback.

3. Feasibility and Viability

Feasibility

A basic version is technically feasible using widely used web-development tools and a relational database. The first version can operate in one local area with a limited number of crops before adding advanced AI. Existing digital agricultural-market initiatives show that online price discovery, buyer access, and logistics coordination are practical areas to build on. The proposal should integrate with existing services where permitted rather than duplicate them.

Potential challenges and risks

- **Digital access and literacy:** Some users may have limited connectivity or difficulty using apps.
- **Data quality:** Inaccurate stock, price, quality, or harvest-date information can reduce trust.
- **Perishable goods:** Delays, poor packaging, or temperature issues may cause spoilage.
- **Demand forecast limitations:** Weather, festivals, sudden price changes, and crop conditions can make forecasts uncertain.
- **Trust, payments, and disputes:** Users need reliable order records, clear cancellation rules, and safe payment handling.
- **Regulatory and operational requirements:** Market rules, food-safety requirements, tax obligations, and payment-provider terms may apply depending on the activity and location.

Strategies to reduce risks

- Provide local-language support, simple screens, and assisted onboarding through FPOs or community partners.
- Use lot IDs, timestamps, stock confirmation, photo uploads, and clear quality/quantity standards.
- Set realistic delivery windows, encourage suitable packaging, and prioritise nearby buyers for highly perishable produce.

- Start with explainable baseline forecasts; display that predictions are estimates, and avoid guaranteeing prices or demand.
- Use secure authentication, minimal personal-data collection, role-based access, encrypted connections, clear dispute procedures, and established payment providers.
- Consult relevant local authorities and domain experts before live commercial deployment.

4. Impact and Benefits

Stakeholder	Expected benefit
Farmers and FPOs	More buyer visibility, improved ability to compare offers, clearer cost information, and possible reduction in avoidable resale layers.
Consumers	More transparent prices and access to produce listings from nearby sellers, with delivery costs shown before ordering.
Retailers and bulk buyers	Easier discovery of suppliers, quantity-based purchasing, and more predictable sourcing.
Transport providers	Potentially better vehicle utilisation through grouped deliveries and planned routes.
Environment and community	Potential reduction in unnecessary travel and food waste if better matching and delivery coordination work as intended.

How success will be measured

- Number of active farmers/FPOs, buyers, and completed orders.
- Average time taken to sell listed produce and percentage of listings sold.
- Farmer net realisation after visible fees and logistics costs, compared with a suitable baseline.
- Buyer total cost and satisfaction compared with their previous sourcing method.
- On-time delivery rate, transport cost per order, and reported spoilage.
- Forecast error over time, evaluated against a simple baseline rather than assumed to be accurate.

Expected results should be validated through a controlled pilot. The platform will not promise a fixed increase in farmer income or a fixed decrease in consumer prices; outcomes depend on local supply, demand, quality, transport, and market conditions.

5. Research and References

The proposal is informed by the following official resources. These are background references for the problem area and potential integration points; FarmLink Direct is a proposed concept, not an existing government platform.

1. National Agriculture Market (e-NAM), Government of India — overview of an electronic trading platform connecting agricultural markets and supporting transparent price discovery.

https://www.enam.gov.in/web/

2. e-NAM Farmers — information on farmer registration, market access, price information, and online payments.

https://www.enam.gov.in/web/stakeholders-Involved/farmers

3. e-NAM Registration Guidelines — includes registration guidance for FPOs/FPCs and information about logistics-provider support.

https://www.enam.gov.in/web/resources/registration-guideline

4. Indian Council of Agricultural Research (ICAR), Kharif Agro-Advisories 2025 — background on agricultural advisories and digital tools used to support farmers.

https://www.icar.org.in/sites/default/files/Circulars/ICAR-En-Kharif-Agro-Advisories-for-Farmers-2025.pdf

Conclusion
FarmLink Direct proposes a practical, phased approach to improve agricultural market connections. By combining transparent listings, direct buyer discovery, shared logistics, and carefully evaluated AI tools, it aims to help farmers make better-informed selling decisions while giving consumers and bulk buyers clearer purchasing options.